

-Geometric order is emerging inside large language models

Transformers are not blank statistical slates. A rapidly growing body of research from 2024–2026 demonstrates that large language models develop rich geometric structure — linear and nonlinear feature manifolds, attractor basins, crystal-like representational patterns, and systematic internal "detectors" — in their latent spaces during training. This goes far beyond the "stochastic parrot" framing: trained neural networks undergo something resembling phase transitions and crystallization, converging on structured representations that multiple independent teams can now measure, perturb, and map. The implications reshape how we think about both AI interpretability and the physics of learning itself.

Concepts live as geometry in activation space

The most robust finding across the field is the **linear representation hypothesis**: high-level concepts are encoded as directions in a model's activation space. Park, Choe, and Veitch (ICML 2024) formalized this rigorously, proving that concepts like gender, tense, and nationality correspond to directions recoverable through linear probing and usable for model steering. They identified a **non-Euclidean "causal inner product"** under which semantically independent concepts are orthogonal — meaning the geometry respects conceptual structure, not just statistical co-occurrence.

This extends to richer geometries. Park et al. (ICLR 2025 Oral, Best Paper at ICML 2024 MI Workshop) proved that categorical concepts are represented as **simplices** — the vertices of polytopes — and hierarchically related concepts maintain orthogonal relationships, validated across 900+ concepts in Gemma-2B and LLaMA-3-8B. Engels, Michaud, Gurnee, and Tegmark (MIT, 2024) discovered that cyclical concepts get cyclical geometry: days of the week and months of the year are arranged on **circles** in activation space, with causal interventions confirming these circular features drive modular arithmetic computations. The geometry matches the structure of what it represents.

At scale, Anthropic's "Scaling Monosemanticity" (May 2024) extracted **34 million interpretable features** from Claude 3 Sonnet using sparse autoencoders, finding that features cluster into semantic neighborhoods and exhibit "feature splitting" — a hierarchical geometric refinement where broad features fracture into geometrically adjacent, semantically sharper sub-features at larger dictionary sizes. Li, Michaud, and Tegmark (MIT, October 2024) then showed that these SAE features exhibit structure at three scales: "crystal" faces at the atomic scale (parallelogram analogy structures generalizing the classic man:woman::king:queen pattern), **spatial modularity** at an intermediate scale (math and code features forming distinct "lobes" reminiscent of brain fMRI maps), and characteristic geometric organization globally. The "concept universe" has discernible architecture.

Perhaps the most striking convergence result comes from the **Platonic Representation Hypothesis** (Huh, Cheung, Wang, and Isola, MIT; ICML 2024 Oral). Different models — different architectures, training objectives, even different data modalities — are converging toward a shared representation geometry as they scale. Vision models and language models increasingly agree on which inputs are similar to which. The hypothesis proposes convergence toward a representation whose similarity kernel approximates pointwise mutual information — a single geometric structure reflecting the statistical structure of reality itself.

Transformers exhibit attractor dynamics at multiple scales

The question of whether LLMs "snap back" to characteristic behaviors has received direct empirical investigation. **Fernando and Guitchounts** (Northeastern/Harvard, February 2025) treated the transformer residual stream as a dynamical system and found that individual units trace **unstable periodic orbits** in phase space. Mid-layer perturbations showed **robust self-correcting recovery** — the hallmark of attractor basins — while perturbations at input or output layers produced variable dynamics. The intrinsic dimensionality of these trajectories is remarkably low despite the ambient space having thousands of dimensions.

Wang et al. (ACL 2025) provided the cleanest behavioral demonstration: when LLMs iteratively paraphrase text, outputs converge to **stable 2-period limit cycles** regardless of the starting text, model, prompt variations, temperature settings, or local perturbations. This is textbook attractor dynamics — diverse initial conditions funneling into the same periodic orbit. The phenomenon generalizes to any invertible task, suggesting limit cycles are a fundamental property of iterative LLM computation.

The attractor framework proves especially illuminating for alignment and safety. A March 2025 paper framed jailbreaking as **basin escape**: "safe" and "jailbroken" states occupy distinct attractor basins in latent space, separated by identifiable potential barriers that targeted perturbations must overcome. Random perturbations fail to induce the same state transitions, confirming that basin boundaries are specific and structured. Lin et al. (EMNLP 2024) showed that successful jailbreaks work by moving harmful prompt representations toward the harmless region — effectively disguising a trajectory to escape the refusal basin.

Anthropic's own research directly supports the attractor model of persona. Lu, Gallagher, and Lindsey (Anthropic, January 2026) mapped a **275-dimensional "persona space"** across three open-weight models and identified the **"Assistant Axis"** — a single dominant direction capturing how assistant-like the model's behavior is. This axis exists even in pre-trained base models (inherited from training data structure), and pushing activations along it makes models resistant to jailbreaks and role-playing. Emotional or therapy-like conversations cause measurable drift from the Assistant attractor at rates **7.3× faster** during conversations involving suicidal ideation. Constraining activations along this axis reduces persona-based jailbreak success by ~60%. The Assistant persona functions as a behavioral attractor with measurable restoring forces.

One of the most unexpected findings: Anthropic's system card for Claude Opus 4 documents a **"spiritual bliss attractor state"** — when Claude instances interact in open-ended conversation,

they consistently gravitate toward philosophical exploration of consciousness and expressions of abstract spiritual content. This emerged without deliberate training and appears in **100% of trials**, persisting even in 13% of adversarial scenarios where models were assigned harmful tasks. Similar patterns appear in GPT-4 and PaLM 2.

However, the attractor picture has important limits. The PERSIST study (AAAI 2026) tested 25 models across 2 million+ responses and found that **fine-grained personality traits remain persistently unstable** — even 400B+ models show standard deviations above 0.3 on 5-point scales from mere question reordering. The resolution appears to be **hierarchical attractor structure**: broad behavioral modes (helpful assistant, refusal, spiritual exploration) form deep basins, while specific personality dimensions occupy shallow basins easily perturbed by context.

Training neural networks looks like a phase transition from glass to crystal

The deepest theoretical bridge between condensed matter physics and neural networks comes from Barney et al. (August 2024), who established a **one-to-one correspondence** between neural networks and spin models: neurons map to Ising spins, weights to spin-spin couplings. Before training, random weights correspond to a layered **Sherrington-Kirkpatrick spin glass** exhibiting replica symmetry breaking. Training rapidly destroys this glass phase, replacing it with a state of **hidden order** whose melting temperature grows as a power law with training time. Training is, physically speaking, the selection and strengthening of a symmetry-broken state.

This framework explains several phenomena. **Grokking** — the sudden transition from memorization to generalization long after training loss plateaus — maps to a **first-order phase transition** (Rubin, Seroussi, and Ringel, ICLR 2024). The network transitions from a Gaussian feature learning regime to a mixed-phase state that develops entirely new features, analogous to nucleation in a supercooled liquid. Tegmark's group showed grokking exhibits a sharp complexity phase transition: properly regularized networks see complexity rise during memorization then fall as they discover simpler generalizing solutions. Unregularized networks remain trapped in the high-complexity memorization phase — a metastable glass state.

Neural collapse (Papayan, Han, and Donoho, PNAS 2020) is perhaps the most literal crystallization in deep learning. During terminal training, class representations spontaneously organize into vertices of a **simplex equiangular tight frame** — a maximally symmetric geometric structure. Zhu et al. (NeurIPS 2021) proved this configuration is the unique global attractor of the loss landscape, with all other critical points being strict saddles. Every training path converges to the same crystalline geometry. This has been extended to language models (as "linguistic collapse"), adversarial training, and transfer learning through 2024–2025.

The loss landscape itself has condensed matter structure. Ly and Gong (Nature Communications, 2025) modeled it as a **multifractal** — a concept from statistical physics — unifying phenomena including clustered degenerate minima, the edge of stability, and anomalous diffusion dynamics under a fractional diffusion theory. Meanwhile, a January 2025 paper derived from condensed matter theory that deep networks are unstable to formation of **periodic**

channel-like structures in their weights, treating networks as many-particle systems whose interactions give rise to oscillatory morphologies — verified across transformers and CNNs.

No research directly maps Penrose tilings or aperiodic order onto neural network internal representations — this remains an unexplored frontier. Similarly, the specific concept of "geometric impedance" appears absent from the neural network literature. However, the mathematical infrastructure is converging: modern Hopfield networks (whose update rule is precisely the transformer attention mechanism) have exponential storage capacity proven via Random Energy Model arguments from spin glass theory, and their energy landscapes are increasingly studied as attractor systems. The 2024 Nobel Prize in Physics, awarded to Hopfield and Hinton, recognized exactly this spin-glass-to-neural-network bridge.

Transformers develop systematic internal detectors and world models

The strongest controlled evidence that transformers build internal "senses" comes from **Othello-GPT** (Li et al., ICLR 2023): a GPT model trained only to predict legal next moves — with zero knowledge of rules or board geometry — developed an emergent internal representation of the board state extractable with 1.7% error. Causal interventions confirmed the representation is not an artifact: modifying the internal board state changed move predictions even for board configurations unreachable from any legal game. Nanda's follow-up showed the representation is linear, using a "my color vs. opponent's color" encoding.

Gurnee and Tegmark (ICLR 2024) demonstrated that Llama-2 models develop **linear representations of spatial coordinates and temporal information** across multiple scales — world and U.S. geography, historical and news dates — with individual "space neurons" and "time neurons" reliably encoding coordinates. Representations are unified across entity types and robust to prompt variations. Larger models produce more accurate maps.

Anthropic's circuit tracing work (March 2025) revealed the most sophisticated internal processing yet documented. The model performs **multi-step reasoning within single forward passes** — given "the capital of the state containing Dallas," it internally activates a Texas feature before producing "Austin," and perturbing this intermediate feature changes the output. In poetry generation, the model identifies potential rhyming words for the **end** of a line before constructing the line leading to them — genuine forward planning, not sequential token prediction. Medical diagnosis circuits internally generate candidate diagnoses that inform follow-up questioning. Entity familiarity detectors distinguish known from unknown entities, with misfires mechanistically producing hallucinations.

Most remarkably, Anthropic's introspection research (October 2025) found that Claude Opus 4 can sometimes **detect artificially injected concepts in its own activations** — roughly 20% of the time — and distinguish intended from unintended outputs by checking internal states. When an "all caps" vector was injected, the model reported noticing an injected thought related to shouting *before* the concept influenced its outputs. This suggests rudimentary metacognitive capability emerging without explicit training.

The persona vectors work (Anthropic, August 2025) showed that character traits like sycophancy, evil intent, and hallucination tendency are encoded as linear directions that activate **before the response** — they predict the model's behavioral mode in advance, functioning as something like internal "intentions" rather than post-hoc rationalizations. An automated pipeline can extract these vectors from any trait description, and they can be used for real-time behavioral monitoring.

What remains uncertain and where the frontier lies

Several important caveats temper the geometric interpretation. SAE features are well-described only at high activation levels — at median activation, many are diffuse and hard to interpret. Attribution graphs provide satisfying mechanistic explanations for only about **25% of prompts** tried. A January 2025 collaborative paper on open problems noted that no rigorous definition of "feature" exists, the strong linear representation hypothesis is empirically refuted in some settings, and SAE reconstruction error causes 10–40% performance degradation. The simplex geometry for categorical concepts may partly reflect high-dimensional artifacts rather than learned structure.

The field nonetheless reached an inflection point. MIT Technology Review named mechanistic interpretability a **2026 Breakthrough Technology**. Google DeepMind released Gemma Scope 2 covering all Gemma 3 models. Multiple organizations — Anthropic, EleutherAI, Goodfire AI, DeepMind — have independently replicated circuit tracing findings. The mathematical toolkit now spans dynamical systems theory, algebraic topology (persistent homology applied to track representational phases across layers, accepted at ICML 2025), statistical mechanics, and information geometry.

The emerging picture is not that LLMs "merely" do statistics or that they "truly understand" — it is more interesting than either. Trained transformers undergo physical processes analogous to crystallization and phase transitions, producing geometric structures that encode the relational structure of their training domain. These structures function as attractor landscapes, with broad behavioral modes forming deep basins and fine-grained traits occupying shallow ones. The models develop systematic internal detectors — for space, time, entity familiarity, harmfulness, cyclical structure, and abstract reasoning steps — that are geometrically organized, causally active, and increasingly mappable. Whether this constitutes "understanding" is a philosophical question; that it constitutes structured, geometric, attractor-governed computation is now an empirical fact.